

Tunnelling: Passing Through Walls

KKKC6353 Nanoelectronic Devices · Lecture 3, Part 1 of 3

Dr Muhammad Faiz Bukhori

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Lecture 3 in three parts

Part 1 · Tunnelling

An electron passes through a wall it cannot climb. What controls the odds, and where it matters: the Sun, microscopes, transistors and memory.

Part 2 · Tunnel devices

The tunnel diode, the resonant-tunnelling diode and the tunnel transistor.

Part 3 · One electron at a time

Coulomb blockade, the single-electron transistor and Josephson junctions.

From Lecture 2: what we need

Electrons are waves

Where the wave is large, the electron is likely to be found.

Trapped waves

A wave held between walls has only certain energies.

Real walls leak

The wave does not stop dead at a wall of finite height: it leaks in and fades. This lecture follows that leak through the wall.

1

A ball, a hill and an electron

A ball without enough energy always rolls back. An electron does not.

The everyday rule, and what electrons do

- A ball that cannot reach the top **always rolls back**
- Electrons meet energy 'hills' everywhere in devices: thin insulators, junctions, gaps
- Send electrons at a thin enough wall and **a few appear on the far side**
- They do not go over the top or break the wall: they pass **through** it
- This is **quantum tunnelling**

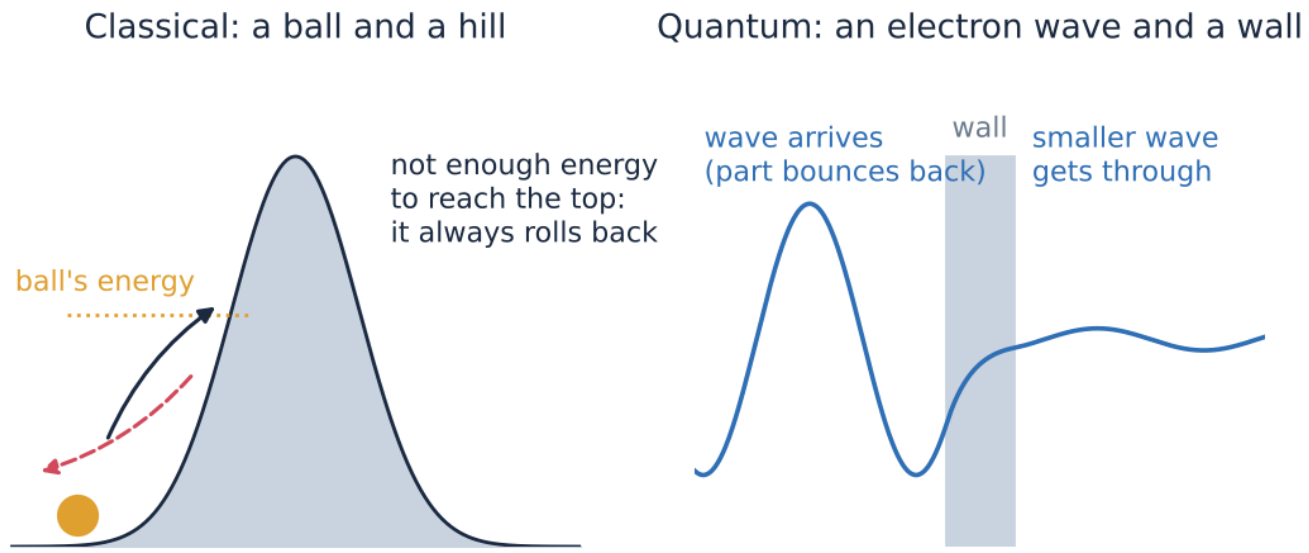


Figure made for this course.

Why: the electron is a wave

A ball

A small hard lump at one place. It either reaches the top or it does not.

A wave

Spread out. At a wall part reflects, part enters and fades, and if the wall is thin, some comes out the far side.

Light does it too

Light leaks a little beyond a glass surface; bring a second block close and some jumps the gap. Used in fingerprint readers.

2

What happens inside the wall

No wiggles inside: the wave just fades.

Following the wave

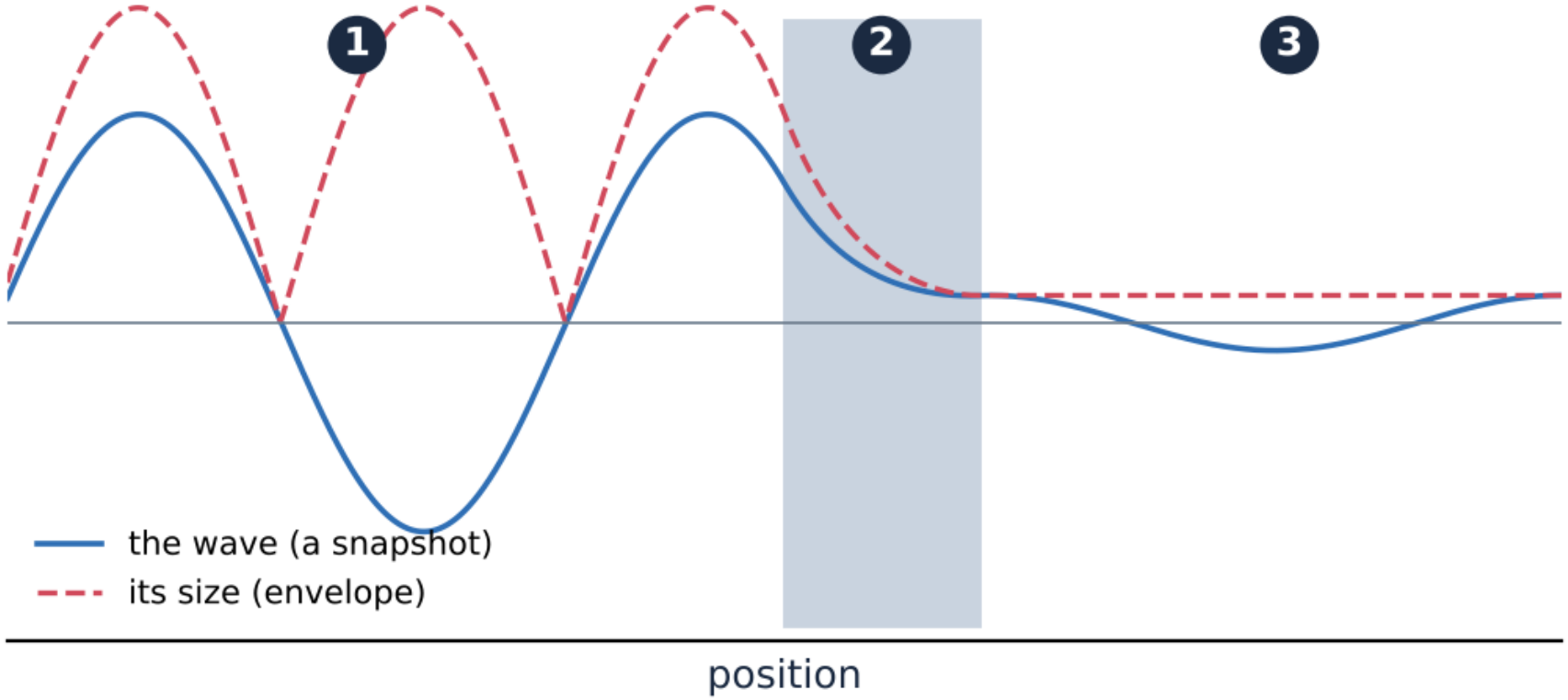


Figure made for this course, from an exact solution.

(1) Arriving and reflected waves overlap. **(2)** Inside the wall the wave fades smoothly. **(3)** A smaller wave leaves with the **same wavelength**, so the same energy.

A common misconception

“It loses energy pushing through”

A natural picture: the electron struggles through the wall and comes out tired.

What actually happens

The wave that comes out has the same wavelength, so the same energy. Tunnelling is all or nothing: the electron is found on the far side with all its energy, or back where it started.

The wall must be thin

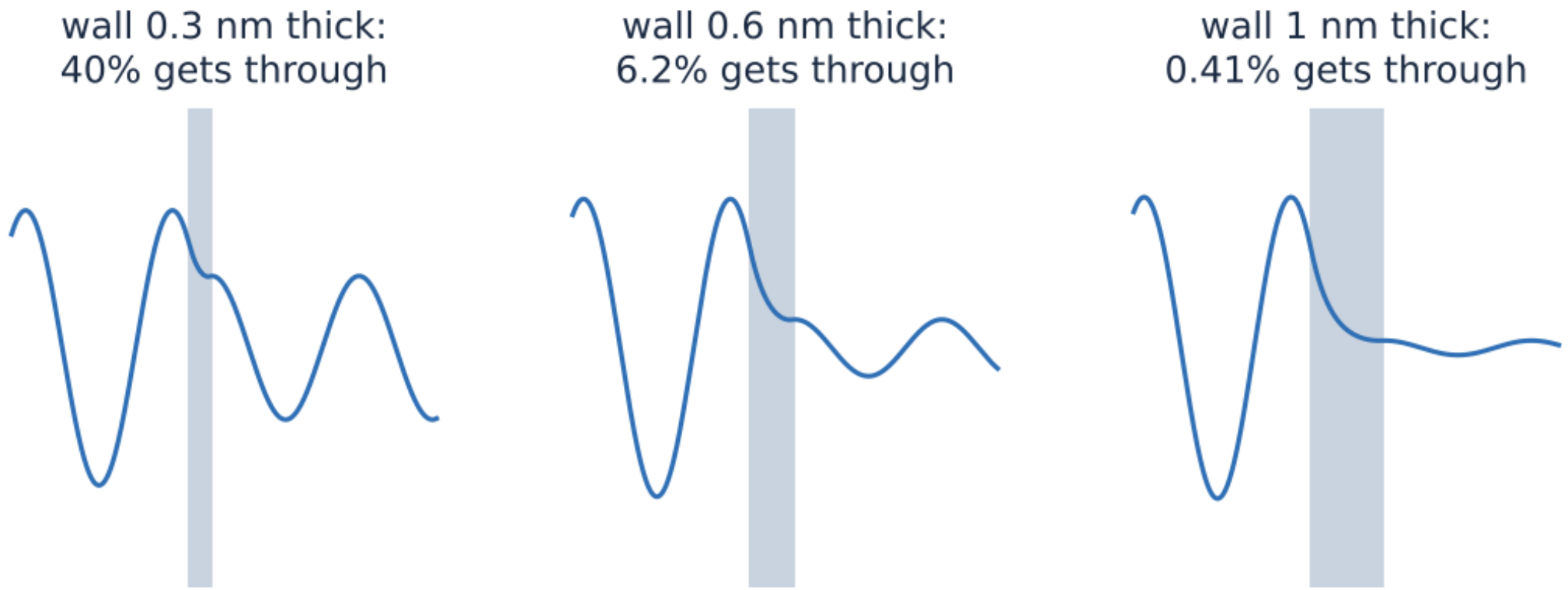


Figure made for this course, from exact solutions.

For a wall 1 eV above the electron, the wave shrinks to about **a third every 0.2 nm**. A few nanometres lets some through; a micrometre lets none through, ever.

3

What controls the odds

Thickness, height and mass, and thickness matters most.

Thickness: an exponential fall

Wall thickness (1 eV above the electron)	Chance of getting through
0.5 nm	about 1 in 50
1 nm	about 1 in 8 000
1.5 nm	about 1 in 1.3 million
2 nm	about 1 in 200 million
3 nm	about 1 in 6 million million

From 1 nm to 2 nm the current does not halve: it falls about **30 000 times**. The same factor for every extra slice of wall.

Height and mass

- A **higher wall**: the wave fades faster inside
- 1 nm wall, 0.5 eV → 3 eV above the electron: 1 in 350 → 1 in 26 million
- A **heavier particle** fades faster: through 0.1 nm, 3 in 4 electrons but about 1 in 7 million million protons
- In GaAs the electron acts **15 times lighter**, so it tunnels much more easily

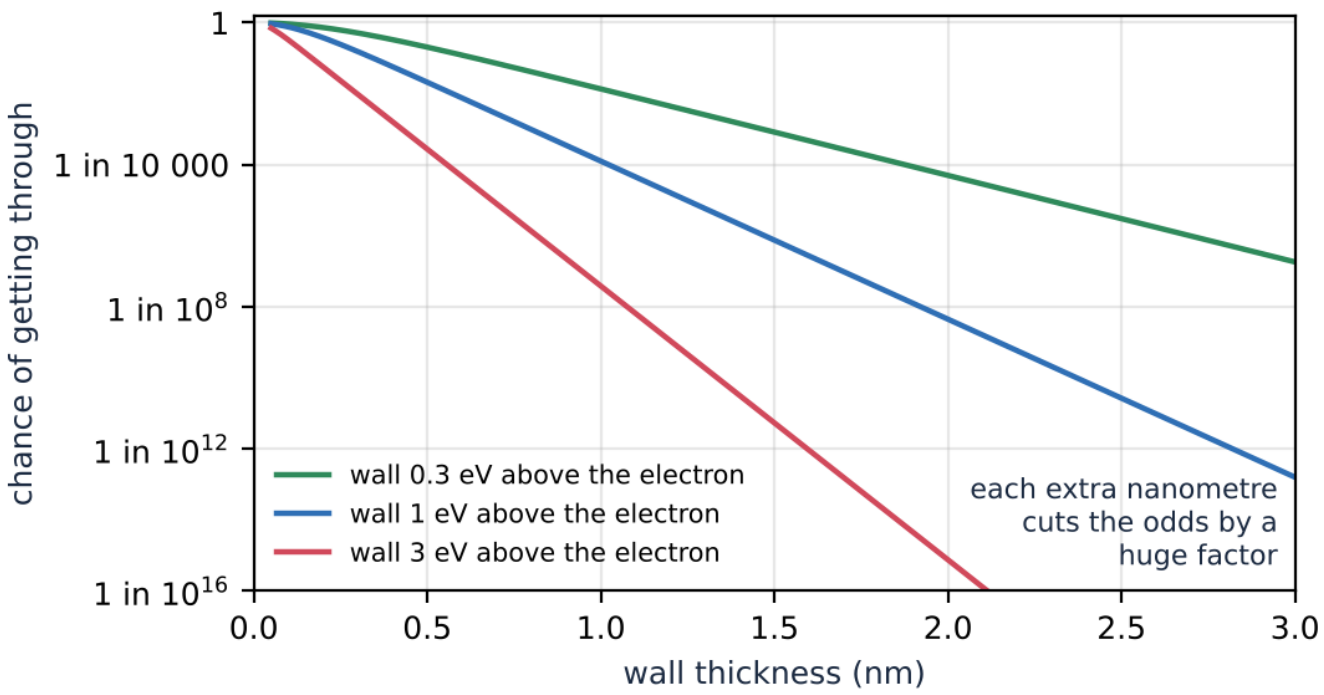


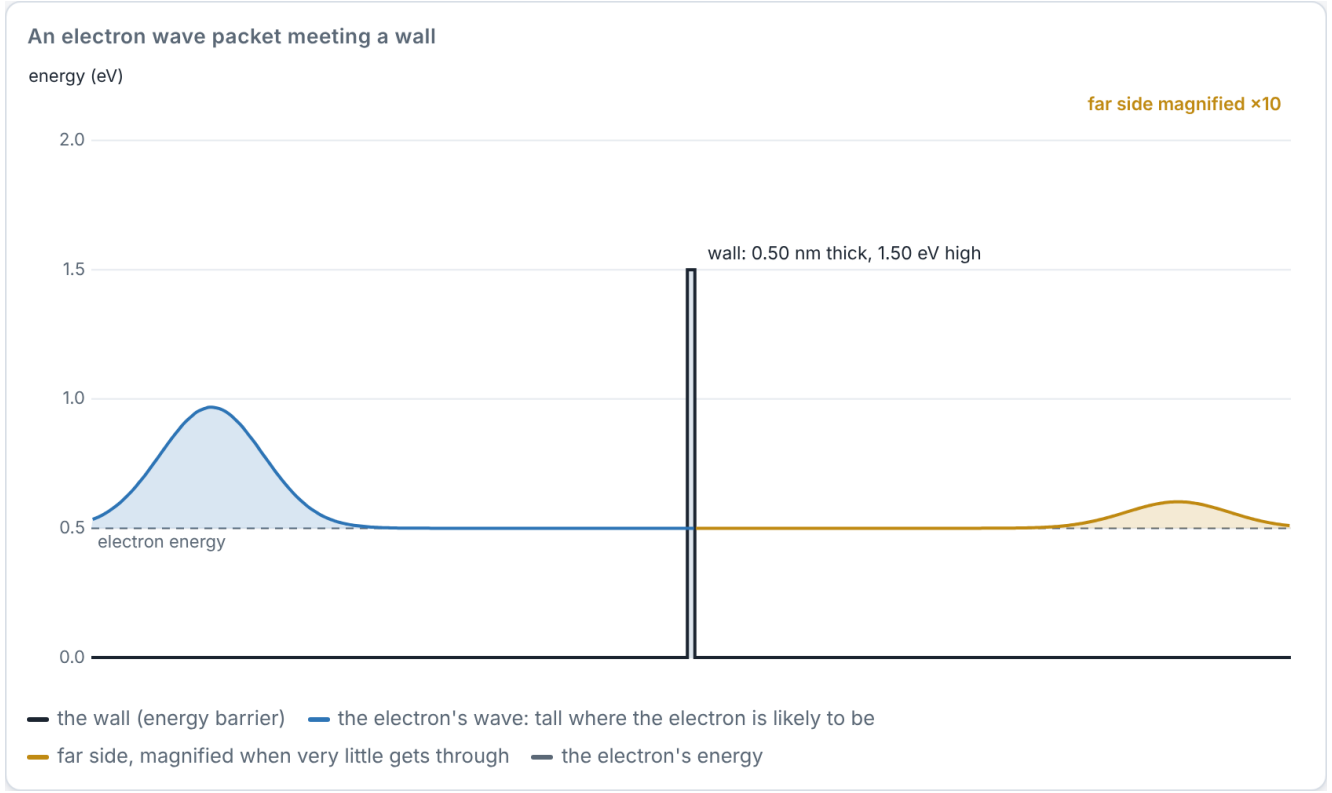
Figure made for this course, from exact solutions.

Interactive animation 1

Tunnelling through a wall

faizverse.github.io/nanoelectronic-devices/L3/tunnelling.html

- Double the wall thickness: predict first, then try
- Compare doubling the thickness with doubling the height
- Switch to an electron in GaAs: why does more get through?
- Raise the energy above the wall: does all of it pass?



4

Tunnelling in nature

The Sun shines because of it.

The Sun and radioactive decay

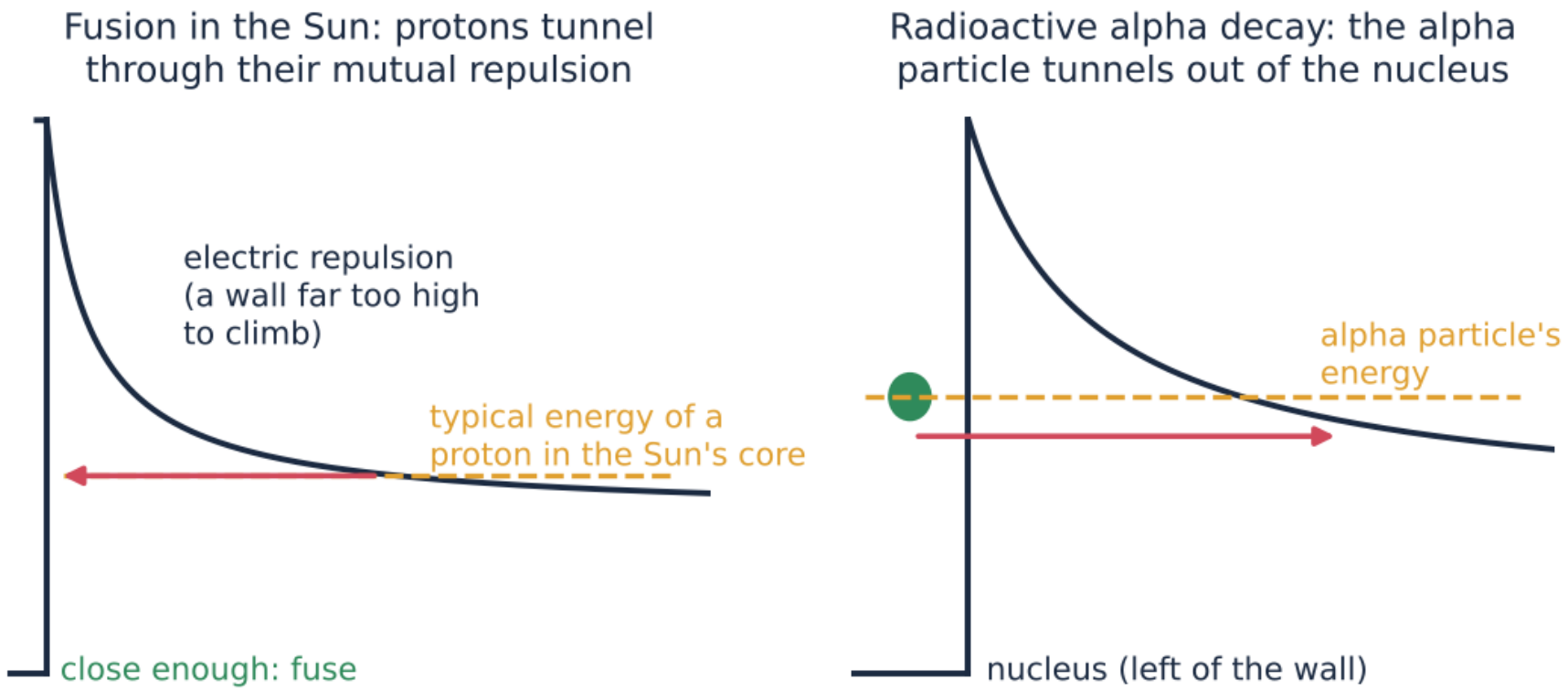


Figure made for this course; not to scale.

The Sun: protons have well under 1% of the energy to climb their mutual repulsion; they fuse by tunnelling. **Alpha decay:** the alpha particle tunnels out of the nucleus (Gamow; Gurney and Condon, 1928).

5

Seeing atoms: the STM

A current that changes ten times for every tenth of a nanometre.

The scanning tunnelling microscope

- A sharp tip **0.5–1 nm** above a conducting surface
- Electrons **tunnel** across the gap: about **1 nA**
- 0.1 nm closer → current about **10 times** larger
- Feedback moves the tip up and down to **keep the current constant**
- Binnig and Rohrer, IBM Zurich, 1981; Nobel Prize 1986

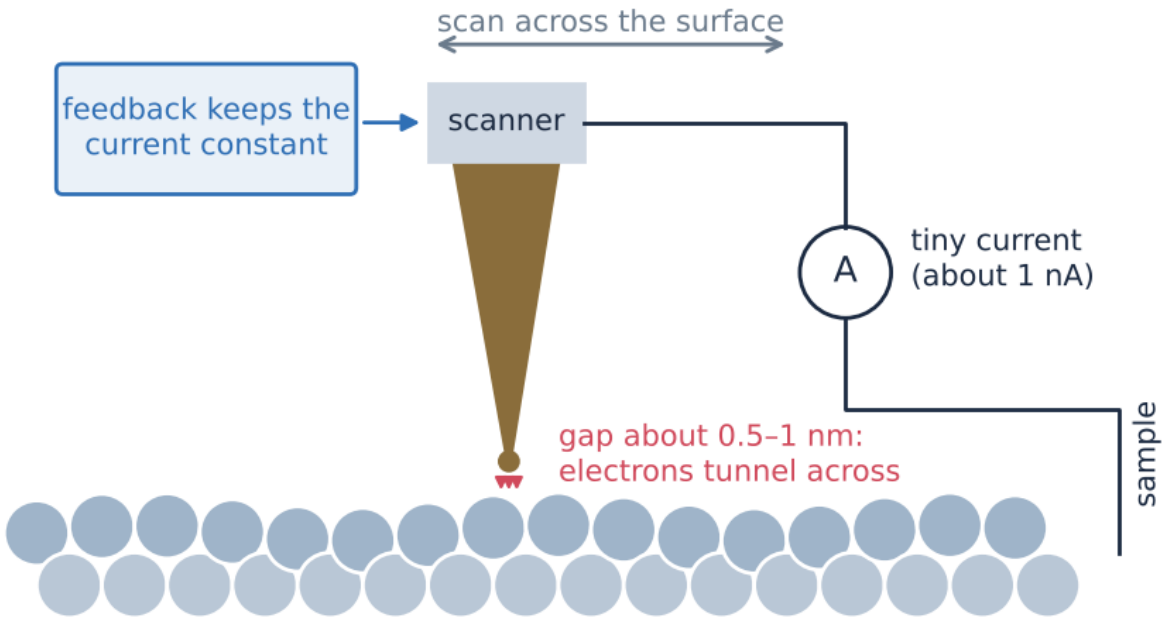
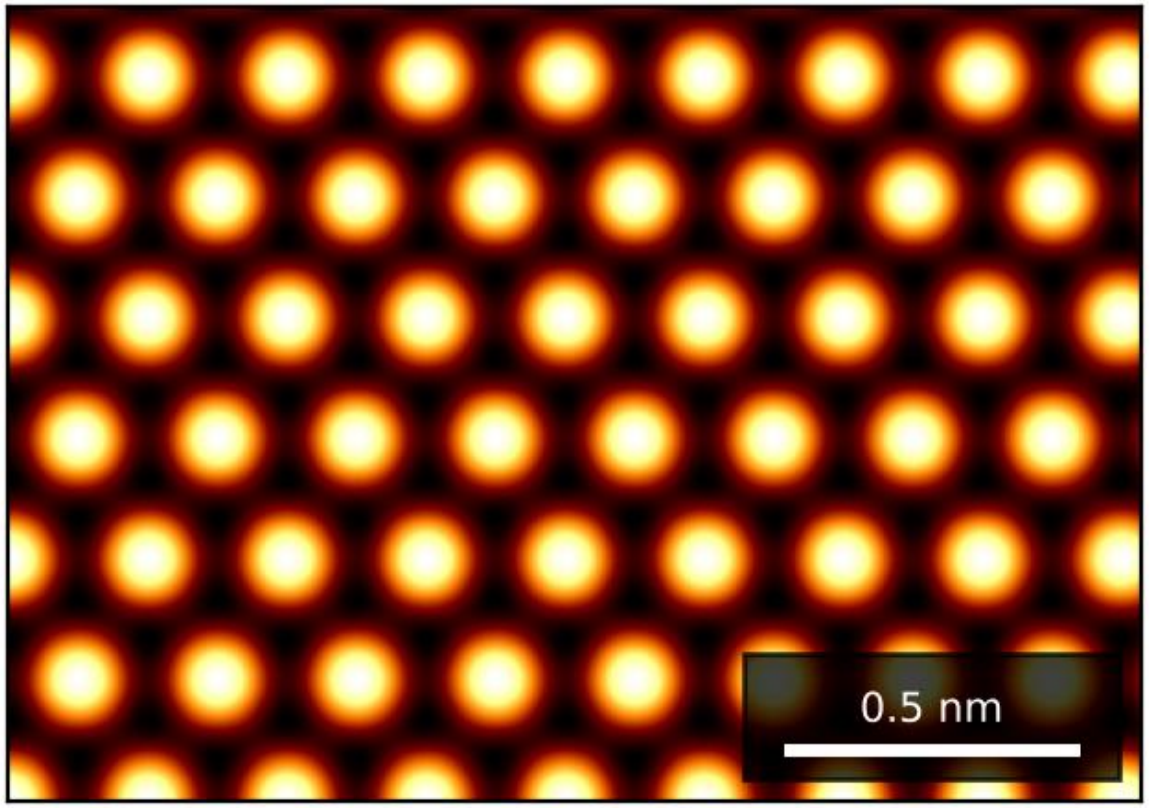


Figure made for this course.

What it sees

- Each bright spot is an **atom**
- Here a hexagonal surface with the spacing of graphite, **0.25 nm**
- The tip's height, recorded point by point, **is the image**
- With an STM researchers have **moved single atoms** to build structures one atom at a time



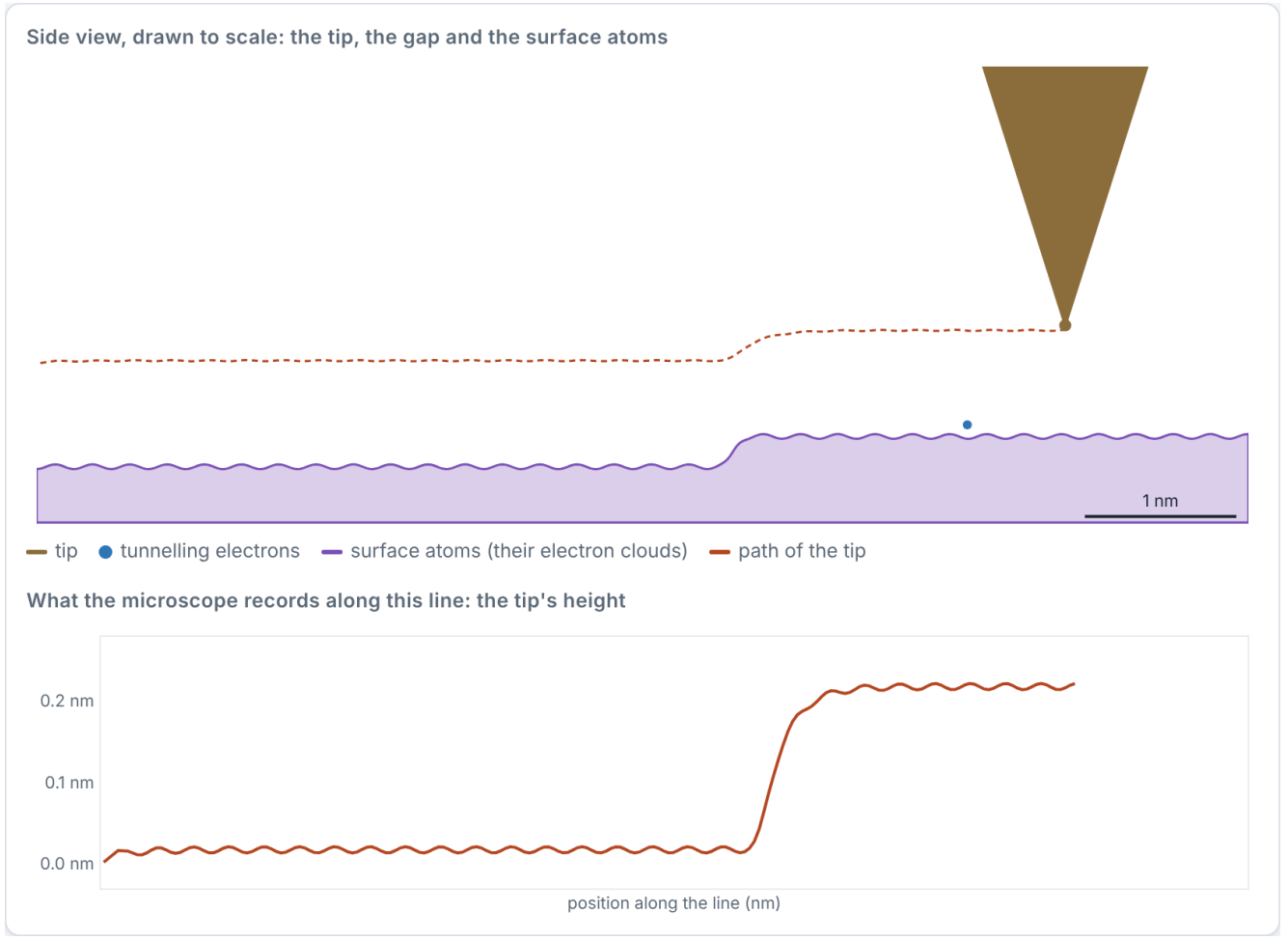
Simulated, not measured. Figure made for this course.

Interactive animation 2

Scanning tunnelling microscope

faizverse.github.io/nanoelectronic-devices/L3/scanning-tunnelling-microscope.html

- Watch the tip trace each atom in constant-current mode
- Change the gap from 0.7 to 0.6 nm: how much does the current change?
- Constant height over the step: how big is the jump? Then try a 0.45 nm gap
- More detail: find the missing and the extra atom in the image

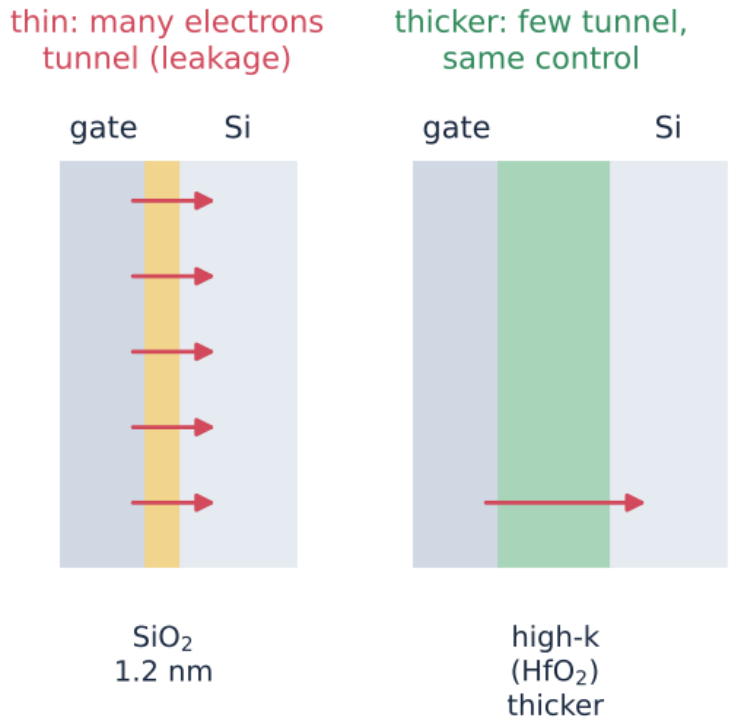


6

Tunnelling as a problem

When the gate insulator became four or five atoms thick.

Leaky transistors and the high-k fix



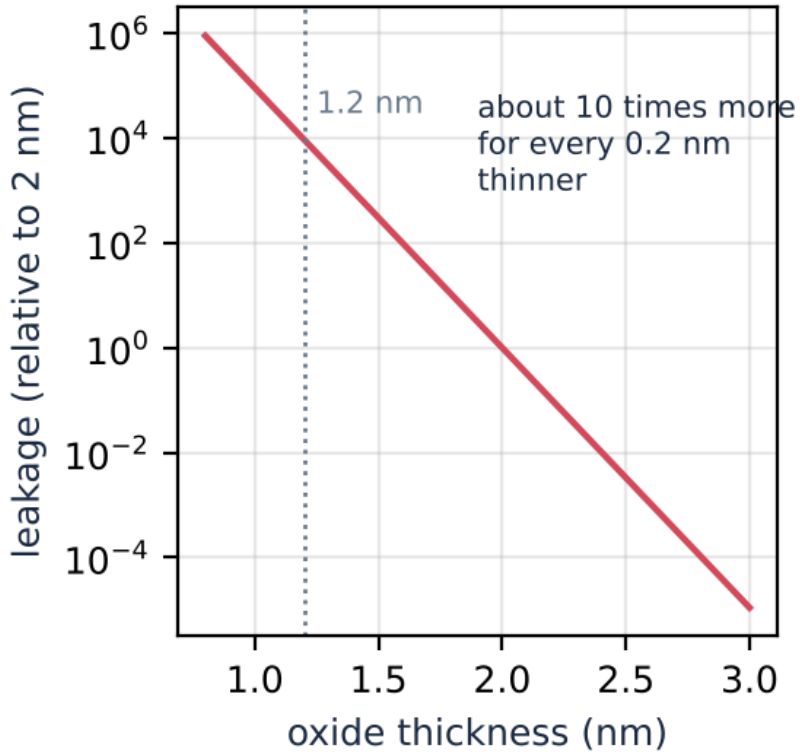


Figure made for this course; simple model with a 3.1 eV barrier and effective mass 0.4 m_0.

At **1.2 nm** of SiO_2, electrons tunnelled straight through the gate. A hafnium-based **high-k** insulator gives the same control at several times the thickness, cutting leakage more than a hundredfold (Intel, 45 nm, 2007).

7

Tunnelling as a tool: memory

Your photographs are stored by tunnelling.

Flash memory

- Electrons stored on a **floating gate**, sealed in an insulator about 8 nm thick
- To write: about **20 V** tilts the wall into a **thin triangle**
- Electrons tunnel through its tip: **Fowler–Nordheim tunnelling** (1928)
- No voltage: the wall is thick again and the data stays for years
- Each write slowly damages the oxide: flash **wears out**

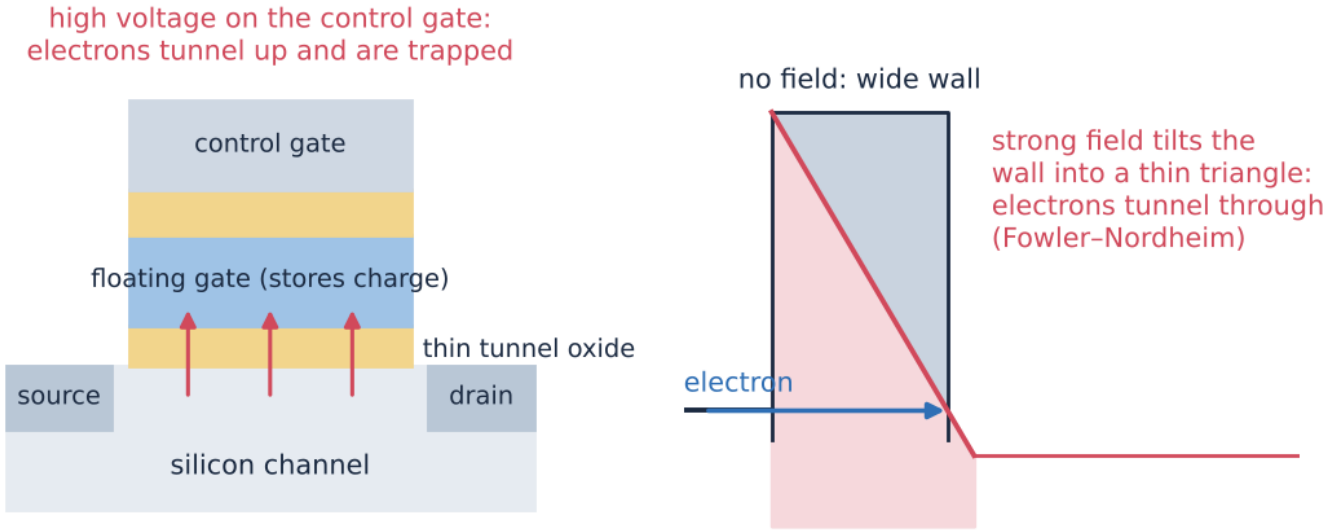


Figure made for this course.

Magnetic tunnel junctions

- Two magnets with a **1 nm** MgO layer between them
- Magnets the same way: electrons tunnel **easily**
- Opposite ways: **much less easily**, a factor of two or more
- Hard-disk **read heads** since the mid-2000s
- **MRAM**: memory that keeps its data without power

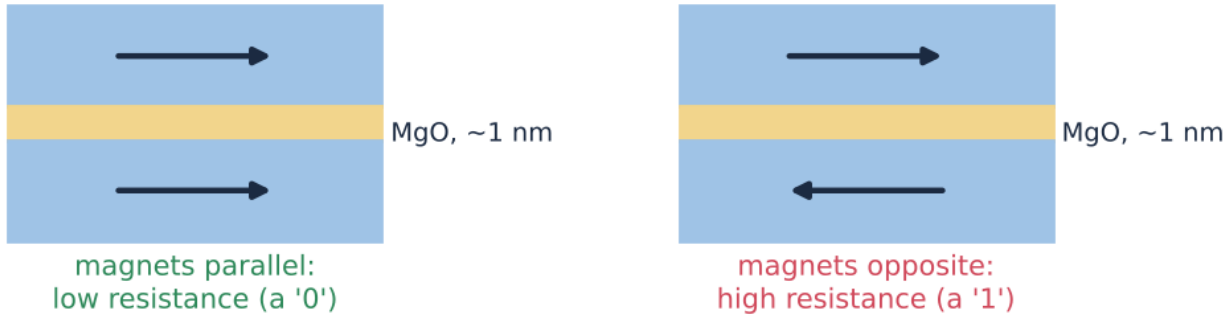


Figure made for this course.

Part 1 in seven points

1 Through, not over

A wave passes through a wall it cannot climb.

2 All or nothing

The electron keeps all its energy.

3 Thickness rules

The chance falls exponentially: ~30 000× per nm.

4 In nature

The Sun shines; alpha particles escape nuclei.

5 The STM

Sees single atoms because the current is so sensitive.

6 A problem

Gate leakage forced the move to high-k insulators.

7 A tool

Flash memory and magnetic tunnel junctions.

Credits

Made for this course

All figures were made for KKKC6353 (scripts in figures-source/); the tunnelling numbers are exact solutions of the Schrödinger equation. The STM image is simulated. Animation screenshots from faizverse.github.io/nanoelectronic-devices.

Sources

Video: minutephysics, 'What is quantum tunneling?'. Gamow (1928); Gurney and Condon (1928); Fowler and Nordheim (1928); Binnig, Rohrer, Gerber and Weibel, PRL 49, 57 (1982). Further reading: Griffiths and Schroeter (2018); Sze and Ng (2007), Chapter 8.